1. Project Title

Propixy

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2. Project Introduction

Propixy is a centralized software platform designed to help property owners, real estate agencies, and facility managers manage residential, commercial, or rental properties more efficiently.

The system streamlines day-to-day operations such as property listing, tenant management, lease tracking, rent collection, maintenance requests, financial reporting, and administrative control. It reduces manual work, improves record accuracy, and provides better visibility into property performance.

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3. Problem the Project Solves

Managing multiple properties manually can lead to:

- scattered tenant and property records
- missed rent payments and lease renewals
- difficulty tracking maintenance issues
- poor communication between tenants, managers, and owners
- lack of real-time reporting on occupancy, income, and expenses

This project solves these issues by offering a single platform to manage all core property operations in an organized and scalable way.

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4. Target Users

- Property Owners who want visibility into property income and operations
- Property Managers responsible for day-to-day property administration
- Real Estate Agencies handling multiple landlords and tenants
- Tenants who need access to rent details, lease information, and maintenance requests
- Admin Staff managing records, reports, and user permissions

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5. Key Features

- Property listing and unit management
- Tenant registration and profile management
- Lease and contract management
- Rent invoicing and payment tracking
- Maintenance request submission and status tracking
- Owner and tenant communication tools
- Occupancy and vacancy monitoring
- Expense and income tracking
- Document storage for leases, IDs, and property files
- Reports and dashboards for financial and operational insights
- Role-based access control for admins, managers, owners, and tenants

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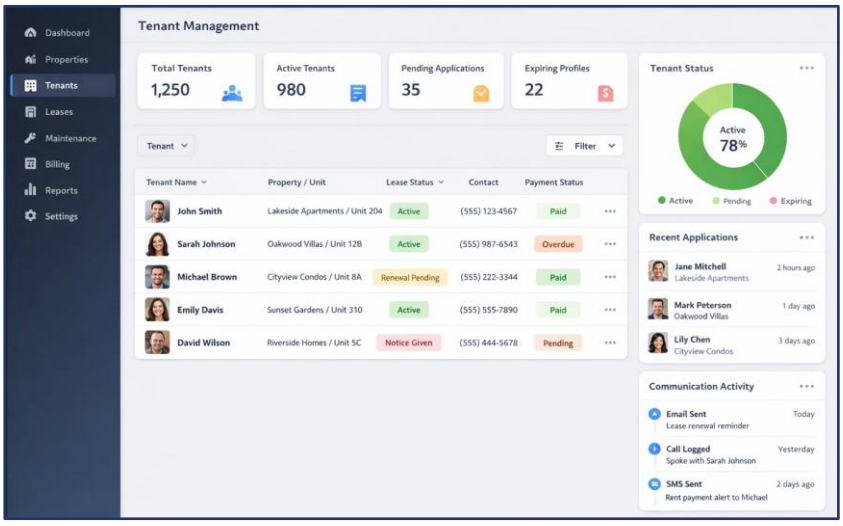
6. System Modules

A. Property Management Module

- Add and manage properties
- Categorize properties by type: apartment, villa, office, shop, building
- Manage units within each property
- Track occupancy status
- Upload property images and documents

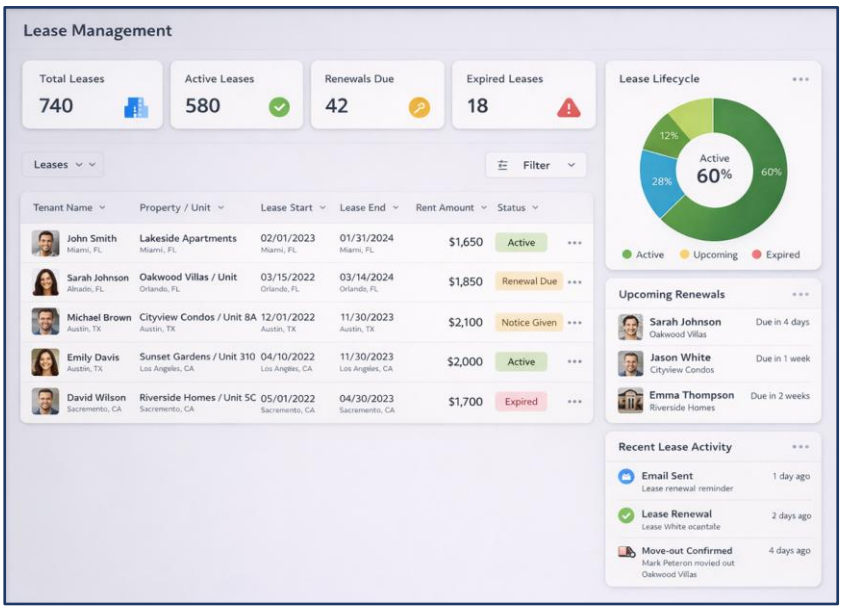
B. Tenant Management Module

- Add tenant profiles
- Store tenant contact information and identification details
- Assign tenants to specific units
- Track move-in and move-out dates
- Maintain rental history



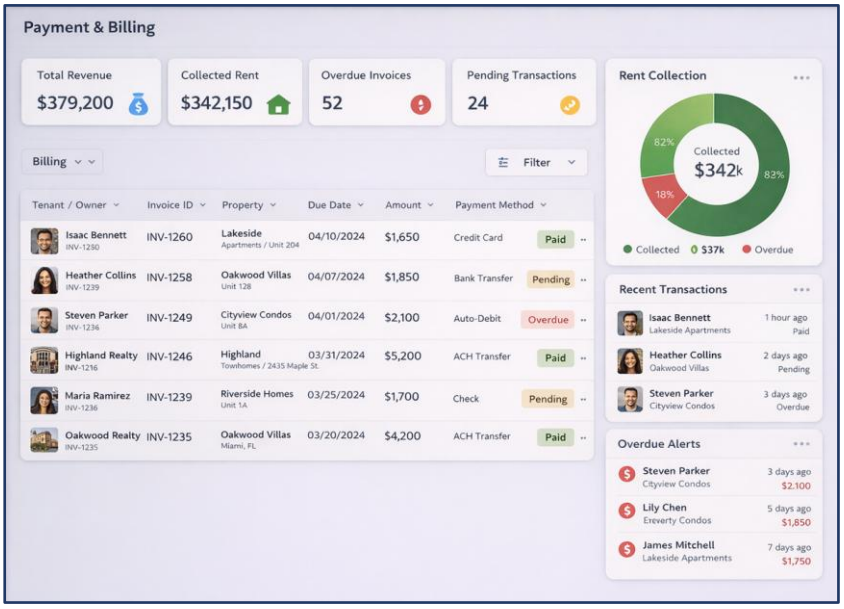
C. Lease Management Module

- Create lease agreements
- Define start date, end date, deposit, rent amount, and payment terms
- Track active, expired, and upcoming renewals
- Upload signed lease documents
- Send renewal reminders



D. Payment & Billing Module

- Generate rent invoices
- Record payments
- Track pending, paid, and overdue rent
- Manage security deposits
- Log penalties or late fees
- View payment history by tenant or property

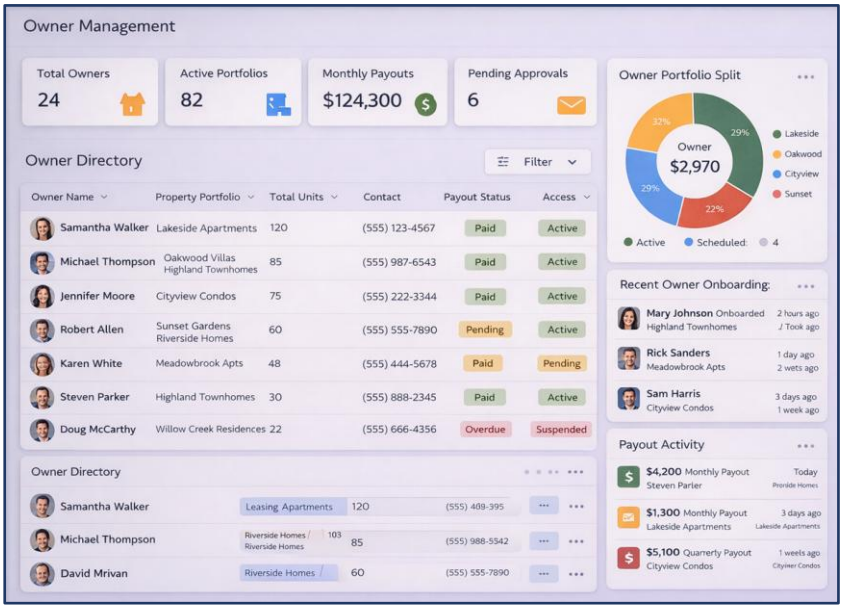


E. Maintenance Management Module

- Raise maintenance tickets
- Assign requests to staff or vendors
- Track status: pending, in progress, completed
- Add notes, cost, and completion date
- Maintain maintenance history per unit/property

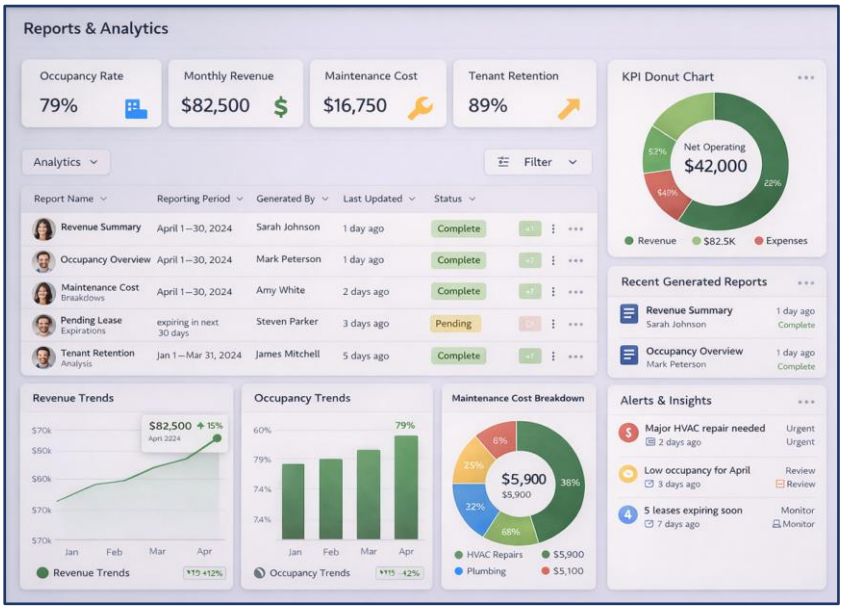
F. Owner Management Module

- Store landlord/owner details
- Map owners to one or multiple properties
- Provide summary of rental income and expenses
- Share reports with property owners



G. Reports & Analytics Module

- Occupancy reports
- Rent collection reports
- Expense summaries
- Property-wise income statements
- Lease expiration reports
- Maintenance cost reports



H. User & Access Control Module

- Admin account management
- Role-based permissions
- User activity tracking
- Secure authentication and password management

7. User Roles

Admin

- Full system control
- Manage users, roles, settings, and overall data
- Access all reports and dashboards

Property Manager

- Manage properties, tenants, leases, payments, and maintenance
- View operational reports
- Coordinate with owners and tenants

Property Owner

- View owned properties
- Monitor rental income, expenses, and occupancy
- Access lease and maintenance summaries

Tenant

- View lease details
- Check rent payment status
- Submit maintenance requests
- Access notifications and documents

Maintenance Staff / Vendor

- View assigned maintenance requests
- Update task progress
- Record service completion details

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8. Core Functional Requirements

- The system must allow admins to create and manage property records.
- The system must support multiple units under a property.
- The system must allow managers to assign tenants to units.
- The system must generate and manage lease agreements.
- The system must track rent schedules, payment status, and overdue balances.
- The system must allow tenants to submit maintenance requests.
- The system must support document uploads for leases and property files.
- The system must generate reports based on rent, occupancy, lease status, and expenses.
- The system must provide role-based dashboards.
- The system must send alerts for overdue rent, lease expiry, and maintenance updates.

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9. Admin Panel Features

- Dashboard with key metrics
    - total properties
    - occupied units
    - vacant units
    - monthly rent collected
    - overdue payments
    - open maintenance requests
- User management
- Role and permission management
- Property and unit approval/control
- Lease tracking overview

- Payment monitoring
- Maintenance oversight
- Report generation and export
- Notification management
- System settings configuration

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10. User Workflow / Journey

Admin Workflow

- Log in to the admin panel
- Add properties and units
- Create user accounts for managers, owners, and staff
- Monitor platform-wide activity and reports

Property Manager Workflow

- Add or update property details
- Register tenants
- Assign tenants to available units
- Create lease agreements
- Generate rent records
- Track payments and overdue accounts
- Handle maintenance requests
- Review reports and property performance

Tenant Workflow

- Log in to tenant dashboard
- View lease and rent information
- Receive rent reminders
- Make payment or view payment status
- Submit maintenance request
- Track request resolution

Owner Workflow

- Log in to owner portal
- View owned properties
- Review occupancy and income
- Check expenses and maintenance summary
- Download statements or reports

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11. Tech Stack

Backend

- ASP.NET Core Web API

Frontend

- React
- Bootstrap

Database

- MS SQL Server

Cloud / DevOps

- Azure
- GitHub CI/CD